** Battery Safety Rules Manual **

RC Car Assembly Manual Series No.0

This manual is the foremost one of our RC car assembly manual series. In this manual, we will explain how to manage the batteries. Please read and follow the instructions carefully. There may be changes in parts due to future discontinuation issues.

RC Car Assembly Manual Series

0. Battery Safety Rules Manual
1. RC Car Hardware Assembly Manual
2. VESC Setup Manual
3. Jetson Xavier NX Setup Manual
4. Keyboard Control with ROS2
5. RC Car Calibration Manual

RLLAB

http://rllab.snu.ac.kr

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1. Introduction

1-1. The battery we will use is a 4-cell Li-Po battery. Each cell has an average voltage of **3.7 V**, giving a total of **14.8 V** when connected in series.

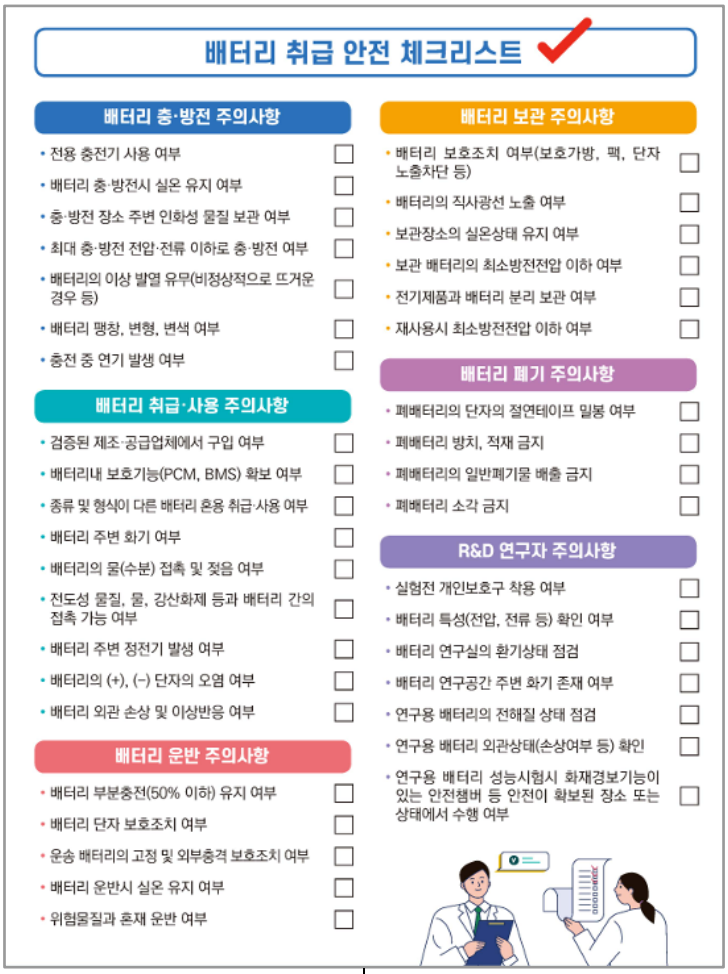


+ [Cell 1] **-** **+** [Cell 2] **-** **+** [Cell 3] **-** **+** [Cell 4] **-**

(Image source : https://www.swaytronic.ch/en/SWAYTRONIC-LiPo-Cell-1S-3.7V-16-000mAh-25C-50C)

1-2. Our battery is not equipped with any safety components such as a Battery Management System (BMS). Therefore, it must be handled with the utmost caution to prevent serious accidents.

1-3. Below is the instruction provided by our university. Please refer to it. In this manual, we will cover how to handle the batteries for this class based on this instruction.



배터리 취급 안전 체크리스트

배터리 충·방전 주의사항
- 전용 충전기 사용 여부 ☐
- 배터리 충·방전시 실온 유지 여부 ☐
- 충·방전 장소 주변 인화성 물질 보관 여부 ☐
- 최대 충·방전 전압·전류 이하로 충·방전 여부 ☐
- 배터리의 이상 발열 유무(비정상적으로 뜨거운 경우 등) ☐
- 배터리 팽창, 변형, 변색 여부 ☐
- 충전 중 연기 발생 여부 ☐

배터리 취급·사용 주의사항
- 검증된 제조·공급업체에서 구입 여부 ☐
- 배터리내 보호기능(PCM, BMS) 확보 여부 ☐
- 종류 및 형식이 다른 배터리 혼용 취급·사용 여부 ☐
- 배터리 주변 화기 여부 ☐
- 배터리의 물(수분) 접촉 및 젖음 여부 ☐
- 전도성 물질, 물, 강산화제 등과 배터리 간의 접촉 가능 여부 ☐
- 배터리 주변 정전기 발생 여부 ☐
- 배터리의 (+), (−) 단자의 오염 여부 ☐
- 배터리 외관 손상 및 이상반응 여부 ☐

배터리 운반 주의사항
- 배터리 부분충전(50% 이하) 유지 여부 ☐
- 배터리 단자 보호조치 여부 ☐
- 운송 배터리의 고정 및 외부충격 보호조치 여부 ☐
- 배터리 운반시 실온 유지 여부 ☐
- 위험물질과 혼재 운반 여부 ☐

배터리 보관 주의사항
- 배터리 보호조치 여부(보호가방, 팩, 단자 노출차단 등) ☐
- 배터리의 직사광선 노출 여부 ☐
- 보관장소의 실온상태 유지 여부 ☐
- 보관 배터리의 최소방전전압 이하 여부 ☐
- 전기제품과 배터리 분리 보관 여부 ☐
- 재사용시 최소방전전압 이하 여부 ☐

배터리 폐기 주의사항
- 폐배터리의 단자의 절연테이프 밀봉 여부 ☐
- 폐배터리 방치, 적재 금지 ☐
- 폐배터리의 일반폐기물 배출 금지 ☐
- 폐배터리 소각 금지 ☐

R&D 연구자 주의사항
- 실험전 개인보호구 착용 여부 ☐
- 배터리 특성(전압, 전류 등) 확인 여부 ☐
- 배터리 연구실의 환기상태 점검 ☐
- 배터리 연구공간 주변 화기 존재 여부 ☐
- 연구용 배터리의 전해질 상태 점검 ☐
- 연구용 배터리 외관상태(손상여부 등) 확인 ☐
- 연구용 배터리 성능시험시 화재경보기능이 있는 안전챔버 등 안전이 확보된 장소 또는 상태에서 수행 여부 ☐

2. RC Car Operation

2-1. Battery cell voltage checker

2-1-1. The voltage checker monitors both the voltage of each cell and the total battery voltage. If the voltage drops too low, it activates a buzzer alarm.
* For safety, always keep the voltage checker mounted while using the battery.





2-1-2. Insert the balance plug into the left side of the voltage checker. The leftmost pin is GND (ground), so be sure to align the black wire of the balance plug with the GND pin.



2-1-3. On the top, there is a small button to set the alarm voltage limit. Press and hold the button until the screen displays **3.3V**. The alarm will sound if any cell's voltage drops below 3.3V.



2-1-3. If the alarm sounds during operation, stop using the battery immediately and recharge it. Always recharge the battery in 'Balance' mode with a current of no more than 2.5A.

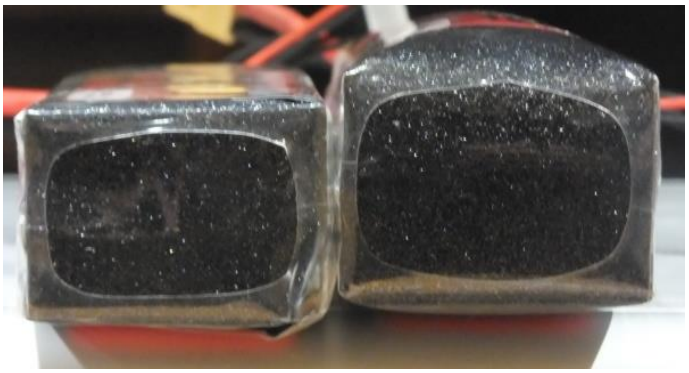
2-2. Cautions

2-2-1. [If the battery is totally discharged]
If any cell's voltage drops below 3.2V, stop using the battery immediately. The battery must be discarded, and you should notify the TAs. We strongly recommend recharging the battery as soon as the alarm sounds.

2-2-2. [If the battery is swollen]
Shut down the power immediately and disconnect all the wires. Then do not touch the battery on your own. Call the TAs.

* Never press a swollen battery—it can damage the internal structure and lead to fire or explosion.

** If you detect smoke or a burning smell, immediately submerge the entire RC Car in the water tank.



(Image source : https://forums.modelflying.co.uk/index.php?/topic/44480-new-lipo-battery-swollen-is-this-normal/)

3. Battery Charging & Storage

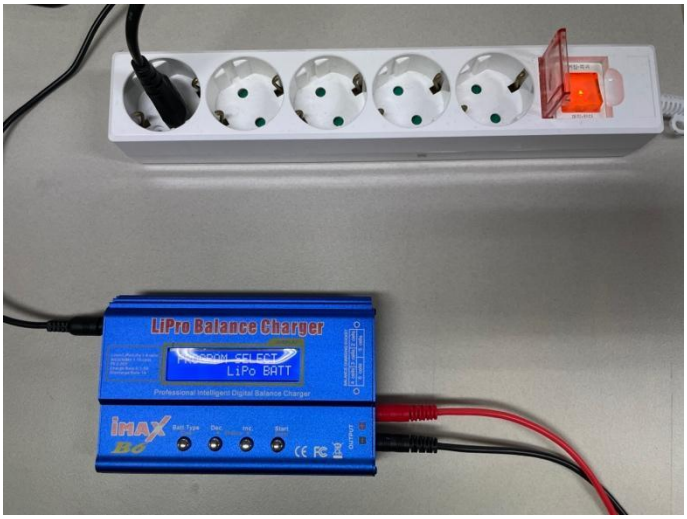
3-1. How to use imax B6 balance charger

3-1-1. Connect banana plug-EC5 cable to imax B6 charger.

* Caution! Never connect banana plug-EC5 cable to the battery first. If the two poles of the banana plug connected to the battery come into contact, it can cause sparks and a short circuit, leading to a serious accident.



3-1-2. Connect the power cable. Set the battery type to “LiPo BATT”.



3-1-3. Connect the EC5 connector of battery to the banana plug-EC5 cable and plug the balance cable of the battery into the charger.



3-1-4. You can see several modes such as “Charge”, “Balance”, “Fast Charge”, “Storage” and so on. Set the mode to “Balance” and adjust the current not more than 2.5A. Adjust voltage setting according to the number of cells in the LiPo battery. In our case, 14.8V (4S)



3-1-5. Press and hold the start (enter) button on the far right to prepare for charging.



3-1-6. The options to cancel or confirm will repeatedly appear. Press the start (enter) button on the far right to begin charging or press the stop button on the far left to cancel.





3-1-7. The battery is now charging. Each cell is calculated to be between 3.8V and 4.2V. Since a 4S LiPo battery is being used, it is adjusted to not exceed the maximum voltage of 16.8V. Once the voltage approaches 16.8V, you should stop charging and the battery is ready for use.



3-1-8. When storing the battery, it should be discharged to the minimum voltage of 3.8V per cell. Change the mode to "Storage", set it to 4S, and start the discharge process in the same manner. Once the discharge is complete, disconnect the battery and store it in a safe place.



* Caution! Never keep LiPo battery without caution, e.g. carrying in your bag, it can be a bomb literally!

3-2. Storage Cabinet

Two safety cabinets have been prepared, and maintenance guidelines have been established for your convenience. Please follow the instructions below.



3-2-1. Each team will be assigned their own area and charger. Batteries must be charged only when you are at your seat. Do not charge the battery when you are away.

3-2-2. Keep the door open while charging the batteries. Make sure to turn off the power once charging is finished.



3-2-3. Keep the recently charged battery in your designated area. Otherwise, place it on the lowest shelf.

※Using the bags is optional — you may decide based on your preference.

4. Emergency Maneuver

4-1. During operation

4-1-1. If thermal runaway occurs (smoke, flames, or odor), stop operation at once and submerge the entire RC car in a water tank. If possible, wear fire-resistant gloves before handling the RC car.





4-1-2. After submerging, make sure to close the lid securely. **Don’t forget to call 119 and evacuate immediately.**

* The gloves must always be kept near the tank.

(Never move them unnecessarily)



4-2. During charging & storage

4-2-1. If a fire is detected by the sensors inside the cabinet, the system will automatically close the door and discharge the extinguishing agent.

4-2-2. The cabinet is designed only to delay the spread of fire. If the door has automatically sealed, evacuate immediately and dial 119.